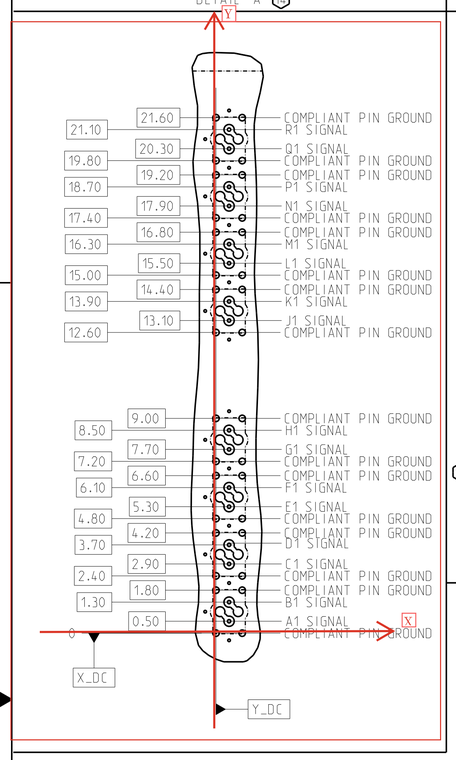
DRW NO.	SH	REV
C-GP820-50009	1	G

ZONE	REV	ECN	DESCRIPTION	BY	DATE	APPROVED
	C	GHEH-25021002	CHANGE COMPONENT PART NUMBERS	HCL-SR	02/10/2025	G.HEATH
	D	GHEH-25021803	CHANGE MODULE PART NUMBER	HCL-SR	02/18/2025	G.HEATH
	F	GHEH-25062002	CHANGED GUIDE PIN MOUNTING HOLE LOCATIONS	HCL-GT	06/20/2025	G.HEATH
	G		ADDED 2D MARKING	HCL-GT	07/30/2025	

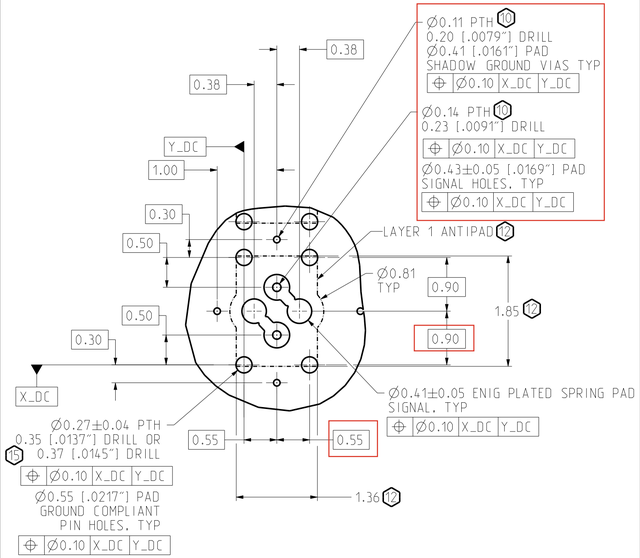
NOTE: OMITTED REVISIONS NOT RELEASED.
CHANGES INCORPORATED IN LATER REVISION.
ONLY THREE MOST RECENT RELEASED REVISIONS DISPLAYED.

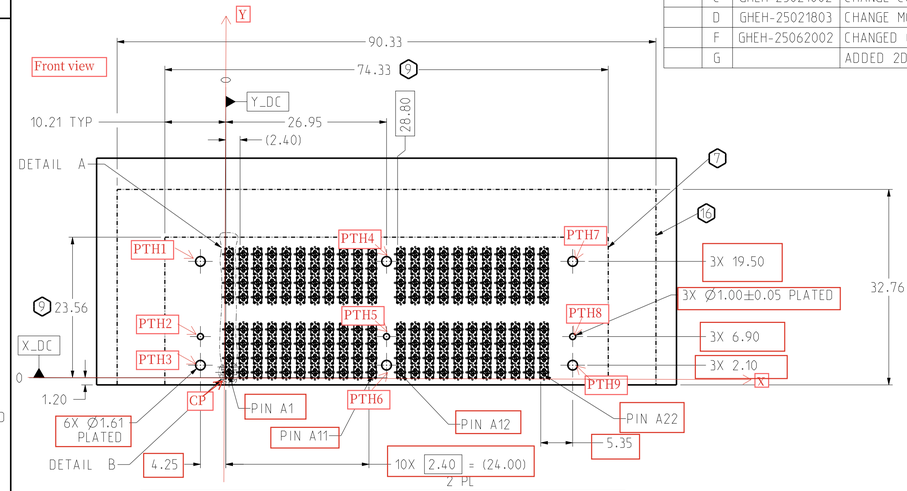
Vendor: AMPHENOL
MPN: GP820-50009
High Speed
Harpoon : "Amphenol_harpoon_foot.pdf "
OLB reference : "Amphenol_GP820-50009_pin_list.xlsx"
Reference "TB2360 - PHD-PHD2 Routing Guide RevD" figure 4 to design exposed copper.

DETAIL 'A' (14)

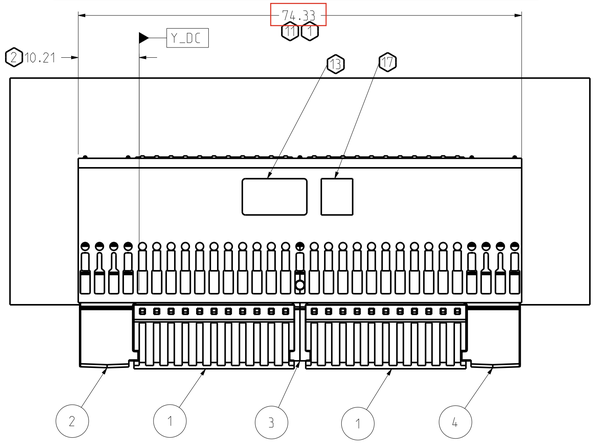


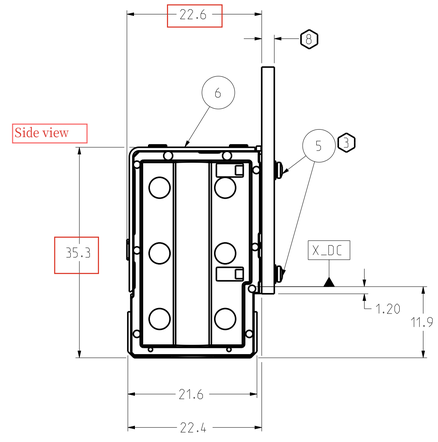
DETAIL 'B' (14)





FRONT VIEW - DC HOLE PATTERN. CONNECTOR SIDE REFERENCED FROM LEFT MOST SIGNAL COLUMN. ALL DIMENSIONS ARE BASIC





FRONT AND END VIEWS

NOTES:

1. TOTAL LENGTH OF STIFFENER.
2. DISTANCE FROM CENTERLINE OF 1ST COLUMN OF GROUND PLATED THRU HOLES TO LEFT EDGE OF CONNECTOR.
3. MOUNTING SCREWS AND JOINER MODULES (IF PRESENT) TO BE BAGGED WITH EACH CONNECTOR ASSEMBLY.
4. INTENDED FOR LAYOUT PURPOSES ONLY.
5. FOR WAFER PITCH USE 2.40MM NOMINAL DAUGHTERCARD LAYOUT
6. SEE TB-2360 (PALADIN HD ROUTING GUIDELINES) FOR BOARD SPECIFIC DETAILS ON ROUTING, PLATING THICKNESSES, SECONDARY DIAMETERS, PAD AND ANTIPAD LOCATIONS, BACKDRILLING AND SHADOW GROUND VIAS.
7. NO SURFACE TRACES IN KEEPOUT ZONE.
8. REFER TO TB-2360 PALADIN HD ROUTING GUIDELINES FOR MIN PCB THICKNESS & MAXIMUM PCB SURFACE FLATNESS IN CONNECTOR PATTERN OF 0.08MM.
9. SURFACE GROUND PLATING MUST EXIST ON PCB TOP LAYER. SEE TB 2360 PALADIN HD ROUTING GUIDELINES FOR TOP AND INNER GROUND LAYER REQUIREMENTS. NO SURFACE TRACES OR OTHER BOARD FEATURES ALLOWED IN CONNECTOR OUTLINE.
10. NO FINISH HOLE TOLERANCE AS HOLE CAN BE PLATED SHUT.
11. OVERALL CONNECTOR LENGTH.
12. SURFACE GROUND PLATING NOT ALLOWED IN LAYER 1 ANTIPAD. SOLDERMASK IS ALLOWED BUT CANNOT OVERLAP SIGNAL PADS.
13. CONNECTOR IS LASER MARKED WITH MANUFACTURING INFORMATION. REFER TO DRAWING C-527-5390-000 FOR MARKING REQUIREMENTS AND DETAILS.
14. DETAIL VIEWS ARE NOT TO SCALE. DO NOT SCALE DRAWING.
15. DRILL SIZE TO BE SELECTED BY PCB FABRICATOR TO REACH FINISHED HOLE SIZE. SEE TB-2360 PALADIN HD ROUTING GUIDELINES.
16. KEEPOUT ZONE FOR CONNECTOR REWORK.
17. REFER TO DRAWING C-527-5390-000 FOR 2D CODE MARKING REQUIREMENTS AND DETAILS.

TABLE 1: PARTS LIST

ITEM	PART NUMBER	DESCRIPTION	QTY
1	658-8QC2-11V	PALADIN HD2, 8PAIR, HSD MOD, 11POS, 92 OHMS, .0145" DRILL, GXT+, STD GOLD	2
2	428-0191-003	PALADIN HD2, 8PAIR, MIMS GUIDE, LEFT	1
3	428-0193-001	PALADIN HD2, 8PAIR, MIMS ENDCAP, MIDDLE, WITHOUT LATCH	1
4	428-0191-004	PALADIN HD2, 8PAIR, MIMS GUIDE, RIGHT	1
(3) 5	809-0011-002	SCREW W/WASHER, #0-80 UNF, 7MM LONG	6
6	617-2816-074	PALADIN HD, STIFFENER	1

NOTE: SEE FRONT VIEW FOR STIFFENER LENGTH.

DRAFT
NOT APPROVED

CUSTOMER: AWS / TETON 2/3 MAX18

TOLERANCES	
0.0	±0.25
0.00	±0.13
0.000	± -
ANGLES	± 3°

INTERPRET PER ASME Y14.5M

CODE IDENT 31413

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN MM. DECIMAL MARKER IS A PERIOD

CUSTOMER USE DRAWING

DESIGN 08/26/2024 G.HEATH
DRAWN 08/26/2024 HCL-GT
CHK 08/26/2024 G.HEATH
APVD 09/11/2024 G.HEATH

Amphenol TCS
A Division of Amphenol Corporation
200 Innovative Way, Nashua, NH 03062 603.879.3000

TITLE: PALADIN HD2 DAUGHTERCARD ASSEMBLY 8 PAIR CONNECTOR, LEADFREE

PART NO. GP820-50009 REV G
DRAWING NO. C-GP820-50009
CREO ASSEM: C-GP820-50009 0.5
DRAWING: C-GP820-50009 0.15

SIZE D | SCALE 2/1 | SHEET 1 OF 1